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LUKE SHIMANUKI

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RECENT EXPERIENCE

Aurora Innovation — **Staff ML Eng** (prom. from Sr. Eng), Behavior ML — 2022–present

- Led 7-engineer team to deliver forecasting models enabling 250k+ incident-free miles of driverless operations.
- Primary technical point of contact for interaction behavior in cross-org launches, demos, and incident reviews; diagnosed on-road failures, led design and delivery of fixes, and communicated timeline and scope trade-offs with senior leadership and external-facing launch/demo teams.
- Proposed and led black-box forecasting validation framework that surfaced previously unnoticed safety-critical failures, prompted mitigations and new test coverage required for driverless launch, and became part of the validation process for every new model release.
- Led scene understanding and off-nominal workstreams for the behavior model, aligning planning, perception, mapping, routing, and validation to ensure safe on-road interactions and fault-response; developed training recipe combining pre-training, LoRA finetuning, and offline-RL/DAgger that resolved a large class of long-tail scenarios, including 100+ emergency-vehicle and fault-handling tests requiring safe pull-over behavior.
- Designed and led new process to rapidly integrate multiple model improvements developed by parallel teams, scaling experiment velocity as the org tripled in size while ensuring a stable, steadily progressing baseline.
- Mentored & sponsored 4 ML engineers, creating opportunities for them to own increasing technical complexity and cross-team/leadership scope by providing design/communication guidance and acting as a safety net to encourage risk-taking; successfully advocated for 1 promotion to senior.

Magna Electronics / Optimus Ride — Senior SDE (prom. from SDE II), Motion Planning — 2020–2022

- Led 3 engineers across 2 teams to expand prediction-planning interface and algorithms to produce probability-aware forecasts and plans, enabling safe autonomous navigation through contested intersections.
- Formulated technical requirements & algorithmic specifications for 2 ADAS features following ISO 26262 functional safety standards, resulting in a patent filing.

EDUCATION & RESEARCH PUBLICATIONS

Massachusetts Institute of Technology | CSAIL Learning & Intelligent Systems — 2016–2020

M.Eng Electrical Engineering & Computer Science (AI Concentration), GPA 5.0 (out of 5.0)

S.B. Computer Science + Brain & Cognitive Science, GPA 4.9 (out of 5.0), 5.0 within CS major

B. Axelrod*, **L. Shimanuki***, "Efficient Motion Planning under Obstacle Uncertainty with Local Dependencies" — WAFR 2022

L. Shimanuki*, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 2 Dimensions" — IJRR 2021

B. Kim*, **L. Shimanuki***, et al., "Representation, learning, and planning algorithms for geometric task and motion planning" — IJRR 2021

B. Kim*, **L. Shimanuki***, "Learning value functions with relational state representations for guiding task-and-motion planning" — CoRL 2019

L. Shimanuki*, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 3 Dimensions" — WAFR 2018

** indicates equal contribution*

SKILLS **Languages:** C, C++, Python **ML/AI:** PyTorch / TRT, TensorFlow, ROS **Tools:** Git, Docker, Linux

EARLY CAREER

Cruise Automation | Intern, Prediction | Summer 2018

- Made forecasts dynamically feasible & intersection-aware, tripling passing tests & reducing disengagements.

Optimus Ride (Self-driving) | SWE Intern, Perception | Summer 2017

- Designed & implemented end-to-end perception-prediction pipeline from object detection through tracking & forecasting, establishing architecture that interfaces between the raw sensor data and the motion planner.

RightHand Robotics (Grasping) | Intern, Machine Learning / Vision | January 2017

- Designed & trained grasping model from RGB-D input, improving pick success rate from cluttered boxes.

Tanius Technology (Prop Trading) | Developer, ML Modeling | 2015–2016

- Built end-to-end ML system for algorithmic trading: data ingestion, model training, & backtesting infrastructure.

ACTIVITIES

Site Manager | *Food for Free COVID-19 Relief Program* | 2020

Directed team of 12 volunteers for packing & delivering groceries to ~300 households weekly.

Program Director, Head Webmaster | *MIT Educational Studies Program* | 2017–2020

Directed educational programs reaching ~3000 MS/HS students, with ~1000 classes taught by ~500 teachers, & run by ~100 volunteers. Mentored 5 future program directors. Maintained website used by ~5000 students.

Software Lead | *AVBotz* | 2012–2016

Led the software team (~12 members) for fully-autonomous submarine for manipulating objects, aiming & shooting torpedoes, & navigating around obstacles. International finalist (7th Place) as a HS team at collegiate RoboSub 2015.

AWARDS

USACO Platinum | **Intel STS 2016 Semifinalist** | **Eagle Scout** | **MIT Battlecode 2018 Finalist**

PROJECTS

C++	Low latency audio streaming to enable remote musicians to play in-sync with under 30ms latency
Python	Musical autocomplete to assist chord & melody composition
Python	Open-source window manager extension enabling grid layout
C	Web browser using Chromium's rendering engine with configurable vi-like key bindings
Java	Neural network AI for a multiplayer platformer fighting game
n/a	Inflatable Sailboat: DIY spritsail rigged to an inflatable raft